

Set Theory for Computer Science

Week 2: Relations

Lists and product sets

Relations and their representations

Inverse and composite relations

Properties of binary relations

Definition and notation of lists

A **list** is an ordered sequence (an *array*) of elements

Examples

DigitList := $\langle 0, 1, 2, \dots, 9 \rangle$

LetterList := $\langle a, b, c, \dots, z \rangle$

MixedList := $\langle t, h, u, \quad, f, e, b, \quad, 1, 1 \rangle$

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LetterList := $\langle a, b, c, \dots, z \rangle$

MixedList := $\langle t, h, u, \text{ }, f, e, b, \text{ }, 1, 1 \rangle$

- ▶ elements in a list have **positions**
- ▶ finite lists have a **length**

$\langle 1, 2 \rangle \neq \langle 2, 1 \rangle$

$\langle 1, 2 \rangle \neq \langle 1, 2, 1 \rangle$

t	h	u		f	e	b		1	1
↑	↑	↑	↑	↑	↑	↑	↑	↑	↑
1	2	3	4	5	6	7	8	9	10

Product sets

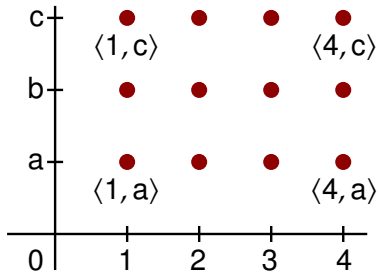
Cartesian product of two sets

$$A \times B := \{\langle a, b \rangle : a \in A \text{ and } b \in B\}$$

$$A \times A =: A^2$$

set of **pairs**
(lists of length 2)

Example



Cartesian product
 $\{1, 2, 3, 4\} \times \{a, b, c\}$

Product sets

Cartesian product of three sets

$$A \times B \times C := \{\langle a, b, c \rangle : a \in A, b \in B, c \in C\}$$

set of **triples**

$$A \times A \times A =: A^3$$

(lists of length 3)

Cartesian product of n sets

$$A_1 \times \cdots \times A_n := \{\langle a_1, \dots, a_n \rangle : a_i \in A_i \text{ for each } i\} \quad (\text{lists of length } n)$$

$$\underbrace{A \times \cdots \times A}_{n \text{ factors}} =: A^n$$

n factors

Product sets

Cartesian product of three sets

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n factors

Examples

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$$

the real plane

$$\text{e.g. } \langle 1, \pi \rangle \in \mathbb{R}^2$$

$$\mathbb{R}^3 = \mathbb{R} \times \mathbb{R} \times \mathbb{R}$$

3d space

$$\text{e.g. } \langle 0, \sqrt{2}, -1 \rangle \in \mathbb{R}^3$$

The product rule

$$\#(A \times B) = \#A \cdot \#B$$

$$\#(A_1 \times \cdots \times A_n) = \#A_1 \cdot \#A_2 \cdots \#A_n \quad \Rightarrow \quad \#(A^n) = (\#A)^n$$

Size of a product set

The product rule

$$\#(A \times B) = \#A \cdot \#B$$

$$\#(A_1 \times \cdots \times A_n) = \#A_1 \cdot \#A_2 \cdots \#A_n \quad \Rightarrow \#(A^n) = (\#A)^n$$

Examples

$$\text{MonthNums} := \{01, 02, 03, \dots, 12\}$$

$$\text{DayNums} := \{01, 02, 03, \dots, 31\} \quad \langle 03, 10 \rangle \quad 3^{\text{rd}} \text{ of October}$$

$$\text{Dates} := \text{DayNums} \times \text{MonthNums} \quad \Rightarrow \# \text{Dates} = 31 \cdot 12 = 372$$

$$\text{Alphabet} := \{a, b, \dots, z\}$$

$$\Rightarrow \#(\text{Alphabet}^4) = 26^4 = 456\,976 \quad \text{number of 4-letter words}$$

Review exercises

Exercise 1

Let $A := \{1, 2\}$ and $B := \{2, 3\}$. Describe the sets $A \times B$ and $B \times A$ explicitly by listing their elements. Is $A \times B$ equal to $B \times A$?

Exercise 2

The number plate of a car consists of a digit, then three letters, and then two digits. Which product of the sets *Digits* and *Letters* lists all possible number plates? How many number plates are there?

Review exercises

Exercise 1

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$$A \times B = \{\langle 1, 2 \rangle, \langle 1, 3 \rangle, \langle 2, 2 \rangle, \langle 2, 3 \rangle\}$$

$$B \times A = \{\langle 2, 1 \rangle, \langle 2, 2 \rangle, \langle 3, 1 \rangle, \langle 3, 2 \rangle\} \neq A \times B$$

Exercise 2

The number plate of a car consists of a digit, then three letters, and then two digits. Which product of the sets *Digits* and *Letters* lists all possible number plates? How many number plates are there?

$$\textit{Digits} \times \textit{Letters} \times \textit{Letters} \times \textit{Letters} \times \textit{Digits} \times \textit{Digits}$$

$$\text{number: } 10 \cdot 26 \cdot 26 \cdot 26 \cdot 10 \cdot 10 = 10^3 \cdot 26^3 = 17\,576\,000$$

Lists and product sets

Relations and their representations

Inverse and composite relations

Properties of binary relations

Relations

A **relation** R of **type** $A_1 \times \cdots \times A_n$ is a subset of $A_1 \times \cdots \times A_n$

Example

Thu, Feb 11, 2016

Time	Platform	Destination	Train
11:00	4	Nijmegen	Intercity
11:03	1	Uitgeest	Sprinter
11:07	4	Rhenen	Sprinter
11:14	1	Schagen	Intercity

Relation of type $Dates \times Times \times Platforms \times Stations \times TrainTypes$

Binary relations and infix notation

A **binary relation** is a relation of type $A \times B$ or A^2

A relation **in the set** A is a binary relation of type A^2

Examples

$IsBrotherOf := \{\langle x, y \rangle : x \text{ is brother of } y\} \subseteq People^2$

$FollowsCourse := \{\langle x, y \rangle : x \text{ follows } y\} \subseteq Students \times Courses$

Binary relations and infix notation

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Alternative infix notation

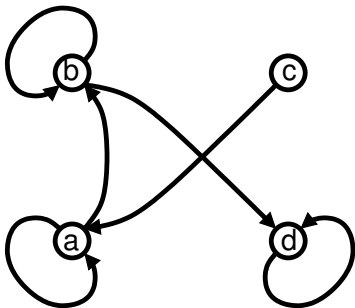
$x R y$ instead of $\langle x, y \rangle \in R$

$x IsBrotherOf y$ instead of $\langle x, y \rangle \in IsBrotherOf$

$x FollowsCourse y$ instead of $\langle x, y \rangle \in FollowsCourse$

Visualization by directed graphs

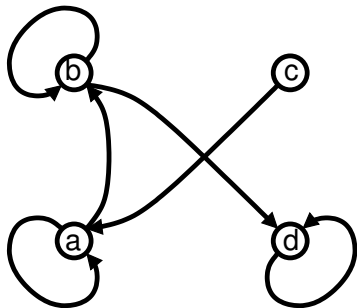
Directed graphs are used to visualize relations in a set A :



$$R = \{\langle a, a \rangle, \langle b, b \rangle, \langle d, d \rangle, \langle a, b \rangle, \langle b, d \rangle, \langle c, a \rangle\} \subseteq \{a, b, c, d\}^2$$

Visualization by directed graphs

Directed graphs are used to visualize relations in a set A :



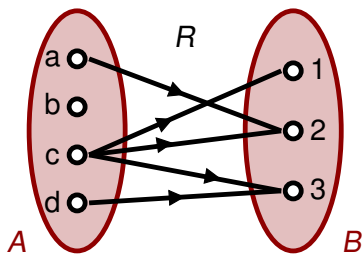
Matrix representation:

	a	b	c	d	(to)
a	1	1	0	0	
b	0	1	0	1	
c	1	0	0	0	
d	0	0	0	1	
(from)					

$$R = \{\langle a, a \rangle, \langle b, b \rangle, \langle d, d \rangle, \langle a, b \rangle, \langle b, d \rangle, \langle c, a \rangle\} \subseteq \{a, b, c, d\}^2$$

Visualization by Venn diagrams

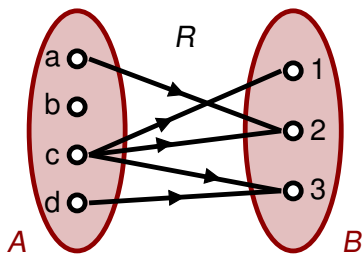
Venn diagrams can be used to visualize all binary relations:



$$R = \{\langle a, 2 \rangle, \langle c, 1 \rangle, \langle c, 2 \rangle, \langle c, 3 \rangle, \langle d, 3 \rangle\} \subseteq \{a, b, c, d\} \times \{1, 2, 3\}$$

Visualization by Venn diagrams

Venn diagrams can be used to visualize all binary relations:



Matrix representation:

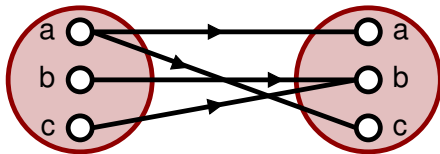
	1	2	3
a	0	1	0
b	0	0	0
c	1	1	1
d	0	0	1

$$R = \{\langle a, 2 \rangle, \langle c, 1 \rangle, \langle c, 2 \rangle, \langle c, 3 \rangle, \langle d, 3 \rangle\} \subseteq \{a, b, c, d\} \times \{1, 2, 3\}$$

Review exercises

Exercise

The following Venn diagram depicts a relation R of type $\{a, b, c\}^2$:

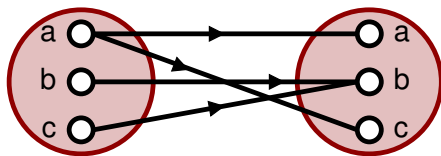


Represent R by a directed graph, a matrix, and as a set of pairs.

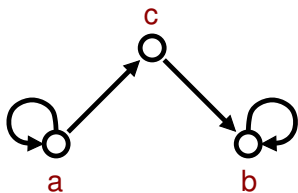
Review exercises

Exercise

The following Venn diagram depicts a relation R of type $\{a,b,c\}^2$:



Represent R by a directed graph, a matrix, and as a set of pairs.



$$\begin{array}{c} a \\ b \\ c \end{array} \begin{bmatrix} a & b & c \\ 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$R = \{\langle a, a \rangle, \langle a, c \rangle, \langle b, b \rangle, \langle c, b \rangle\}$$

break

Lists and product sets

Relations and their representations

Inverse and composite relations

Properties of binary relations

Inverse of a binary relation

The **inverse** of a relation R is $R^{-1} := \{\langle x, y \rangle : \langle y, x \rangle \in R\}$

$$\text{NB: } R \subseteq A \times B \Rightarrow R^{-1} \subseteq B \times A$$

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Example

LivesIn
has type $\text{People} \times \text{Cities}$

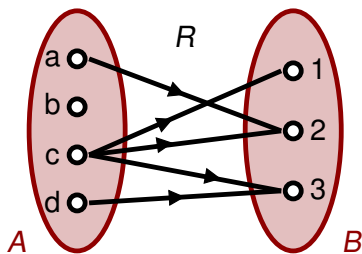
$(\text{LivesIn})^{-1} = \text{IsResidenceOf}$
has type $\text{Cities} \times \text{People}$

$\langle \text{Jan}, \text{Amsterdam} \rangle \in \text{LivesIn}$
Jan LivesIn Amsterdam

$\langle \text{Amsterdam}, \text{Jan} \rangle \in (\text{LivesIn})^{-1}$
Amsterdam IsResidenceOf Jan

The inverse in different representations

Inverting a relation = reversing the arrows:



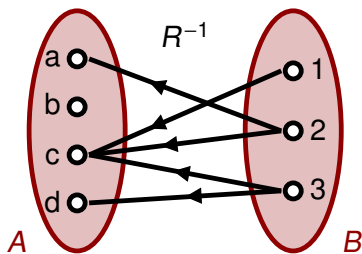
Matrix representation:

	1	2	3
a	0	1	0
b	0	0	0
c	1	1	1
d	0	0	1

$$R = \{\langle a, 2 \rangle, \langle c, 1 \rangle, \langle c, 2 \rangle, \langle c, 3 \rangle, \langle d, 3 \rangle\} \subseteq \{a, b, c, d\} \times \{1, 2, 3\}$$

The inverse in different representations

Inverting a relation = reversing the arrows:



Matrix representation:

	a	b	c	d
1	0	0	1	0
2	1	0	1	0
3	0	0	1	1

(transposed matrix)

$$R^{-1} = \{\langle 2, a \rangle, \langle 1, c \rangle, \langle 2, c \rangle, \langle 3, c \rangle, \langle 3, d \rangle\} \subseteq \{1, 2, 3\} \times \{a, b, c, d\}$$

Composition of binary relations

The **composition** of R and S (i.e. R **after** S) is the relation

$$R \circ S := \{\langle x, z \rangle : x S y \text{ and } y R z \text{ for some } y\}$$

NB: $S \subseteq A \times B, R \subseteq B \times C \Rightarrow R \circ S \subseteq A \times C$

Composition of binary relations

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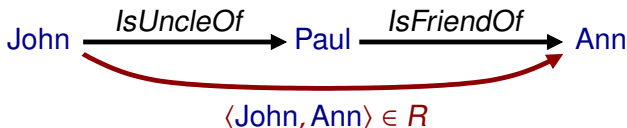
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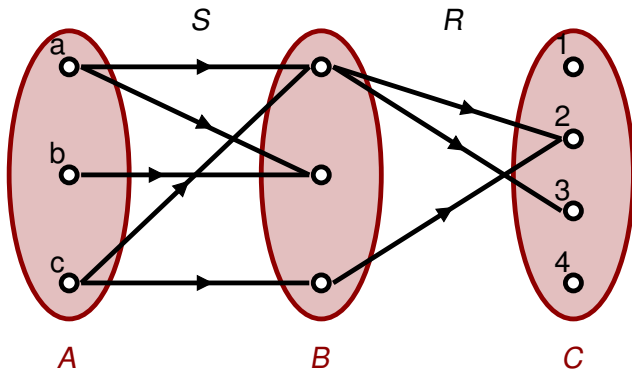
Example

$$R := \text{IsFriendOf} \circ \text{IsUncleOf}$$

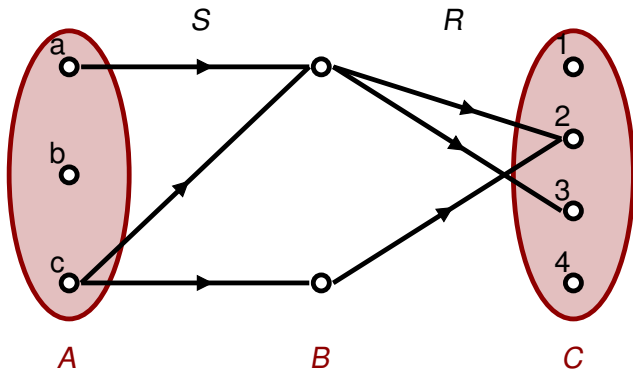
$$\Rightarrow R = \{\langle x, z \rangle : x \text{ is an uncle of a friend of } z\} \quad \text{(note the order!)}$$



Finding the composition using Venn diagrams



Finding the composition using Venn diagrams



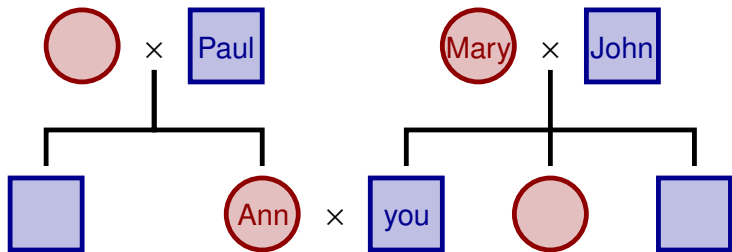
$$R \circ S = \{\langle a, 2 \rangle, \langle a, 3 \rangle, \langle c, 2 \rangle, \langle c, 3 \rangle\}$$

Order of composition

The order of composition matters:

$IsMarriedTo \circ IsParentOf = IsParentInLawOf$

$IsParentOf \circ IsMarriedTo = IsParentOf$



Exercise

In the set of *People*, consider the relations

$IsParentOf := \{\langle x, y \rangle : x \text{ is parent of } y\}$

$IsSiblingOf := \{\langle x, y \rangle : x \text{ is brother or sister of } y\}$

- (a) What is the interpretation of the inverse of *IsParentOf* ?
- (b) What is the interpretation of $(IsParentOf \circ IsParentOf)^{-1}$?
- (c) What is the interpretation of $(IsParentOf \circ IsSiblingOf)^{-1}$?

Exercise

In the set of *People*, consider the relations

$IsParentOf := \{\langle x, y \rangle : x \text{ is parent of } y\}$

$IsSiblingOf := \{\langle x, y \rangle : x \text{ is brother or sister of } y\}$

(a) What is the interpretation of the inverse of *IsParentOf* ?

$$(IsParentOf)^{-1} = IsChildOf$$

(b) What is the interpretation of $(IsParentOf \circ IsParentOf)^{-1}$?

$$(IsParentOf \circ IsParentOf)^{-1} = IsGrandChildOf$$

(c) What is the interpretation of $(IsParentOf \circ IsSiblingOf)^{-1}$?

$$(IsParentOf \circ IsSiblingOf)^{-1} = IsNieceOrNephewOf$$

(not by marriage)

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Inverse and composite relations

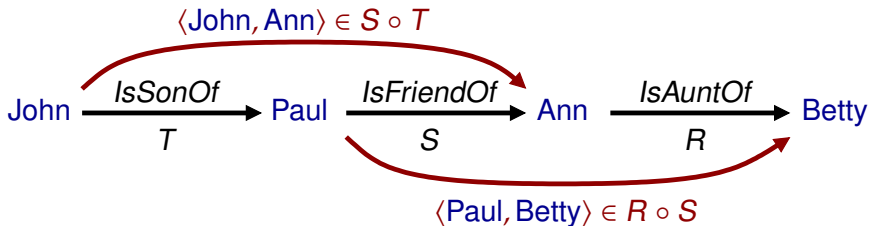
Properties of binary relations

Associativity of composition

Composition is associative: $(R \circ S) \circ T = R \circ (S \circ T)$

Example

Relation $\{ \langle x, y \rangle : x \text{ is a son of a friend of an aunt of } y \}$



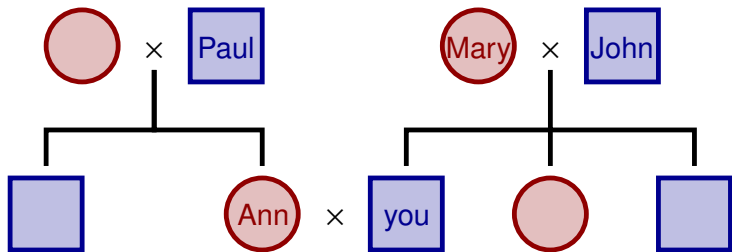
Inverse of a composite relation

The inverse of $R \circ S$ is $(R \circ S)^{-1} = S^{-1} \circ R^{-1}$ (note the order!)

Example

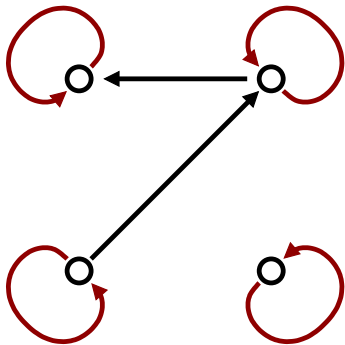
$IsParentInLawOf = IsMarriedTo \circ IsParentOf$

$IsChildInLawOf = IsChildOf \circ IsMarriedTo$



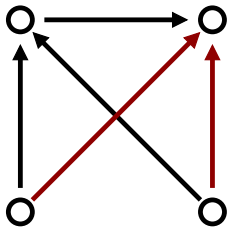
Properties specific for relations *in* a set A

Reflexivity



for all elements x :
 xRx

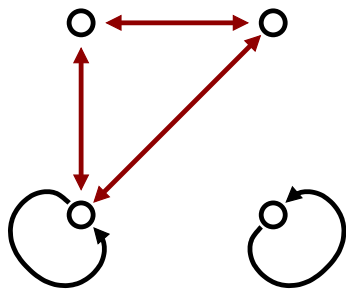
Transitivity



for all elements x, y, z :
 $xRy \wedge yRz \rightarrow xRz$

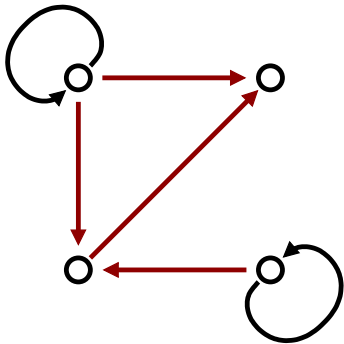
Properties specific for relations *in* a set *A*

Symmetry



for all elements x, y :
 $x R y \rightarrow y R x$

Anti-symmetry



for all elements $x \neq y$:
 $x R y \rightarrow \neg y R x$

Review exercises

Exercise

For each of the following relations in the set of *People*, strike through those properties that the relation does *not* (necessarily) have:

(a) *IsFriendOf*

reflexive transitive symmetric anti-symmetric

(b) *IsParentOf*

reflexive transitive symmetric anti-symmetric

(c) *IsDescendentOf*

reflexive transitive symmetric anti-symmetric

Review exercises

Exercise

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(a) *IsFriendOf*

~~reflexive~~ ~~transitive~~ symmetric ~~anti-symmetric~~

(b) *IsParentOf*

~~reflexive~~ ~~transitive~~ ~~symmetric~~ anti-symmetric

(c) *IsDescendentOf*

~~reflexive~~ transitive ~~symmetric~~ anti-symmetric